REPORT 02 / BUILD / 12 NOVEMBER 2026

CARD ORBIT BUILT.

The Scroll Hero Frame is built: scroll-scrubbed subject, live lit image-slot cards flowing around it, the full text layer, desktop and mobile layouts, and a finish modelled on the D:\day engine.

FOLDER	REF_httpsgrubogrubo_com/scroll-hero-frame
RUN	node tools/dev-server.mjs -> http://localhost:5174
TESTED	1600 x 832 desktop, 375 x 812 phone, in the app's browser
STATUS	Working with placeholder subject + rupee cards. Needs your real subject footage.

1. What was built

The frame now has two layers working together. The **text layer** (headline, lead, glass profile card, awards, signature) is the measured replica from Report 01. Behind it, a **WebGL2 Card Orbit layer** draws the subject video and dozens of lit photo-print cards that flow around the subject as you scroll, passing both in front of and behind it.

Your three requests, and how they were met

You asked for	Built as
All the lighting effects on the cards that flow around the subject	Per-card warm key light + cool fill, glossy sheen with sweeping glints, fresnel rim, golden edge glow on the white border, depth fog, depth-of-field blur; golden light ribbons and bokeh dust; bloom, filmic tone curve and grade over everything.
Non-robotic flow, like the video	Scroll is damped (inertia), fast scrolls add a 'swirl' that settles back, every card drifts and tumbles on its own noise curves, the paper flexes, and the camera has a slow push-in plus hand-held drift. Slow ambient motion continues when you stop.
Cards as empty image slots, filled with random rupee images for now	Every <img> inside .shf-cards is a card face (12 images fill 34 cards). Placeholder rupee SPECIMEN art is in assets/cards/. With no images the cards show empty glass slots.
Use D:\day for flow and texturing	Ported its post chain (soft-knee 13-tap bloom prefilter, dual-Kawase blur, filmic toe/shoulder, teal/cream grade, mid-tone grain, triangular dither), its gradient-noise fbm, and its time-pure, exponentially damped motion model.

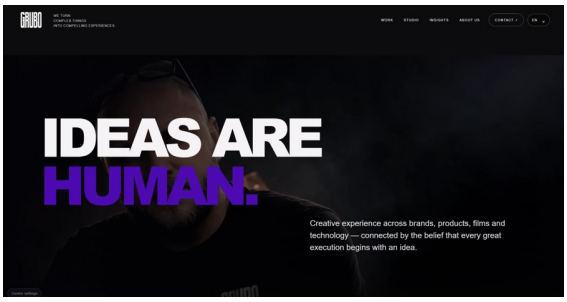
Files

```
scroll-hero-frame/
  index.html               demo page (header + frame + spacers)
  scroll-hero.css          styles; + canvas layer and packed-video rules
  scroll-hero.js           scrub, progress bar, reveals; publishes section._shf.progress()
  scroll-hero-cards.js     NEW  WebGL2 Card Orbit engine (no dependencies, ~600 lines)
  README.md                usage, options, video encoding
  assets/scroll.mp4        placeholder subject, packed colour | matte, 2560x720, keyframe every frame
  assets/poster.jpg        first colour frame
  assets/cards/rupee-01..12.jpg   placeholder card images
  tools/dev-server.mjs     local server with Range requests (video seeking)
  tools/make_placeholders.py      regenerates placeholder cards and subject
```

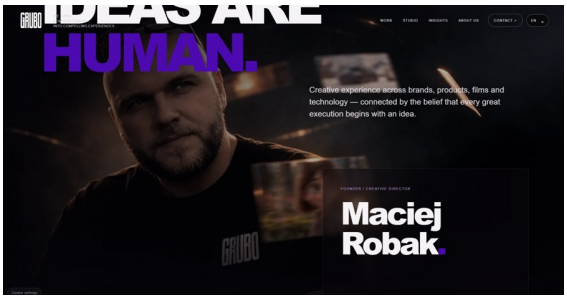
2. Reference vs build

Same 1600 x 832 view and the same scroll positions as your reference screenshots. The subject is a placeholder silhouette until your own footage goes in; everything else is the live build.

Reference



33%: section enters, headline + lead



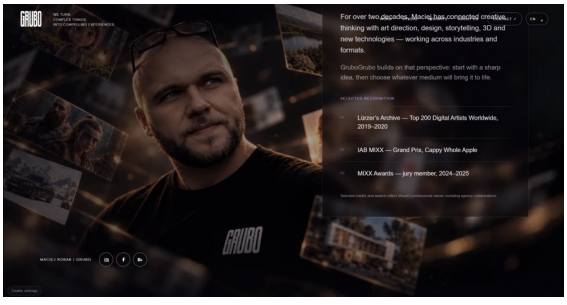
52%: headline leaving, card arriving



61%: counter in, cards gathering

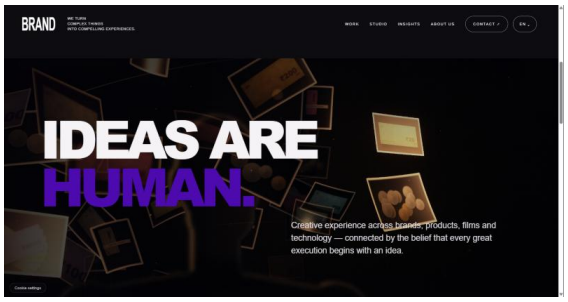


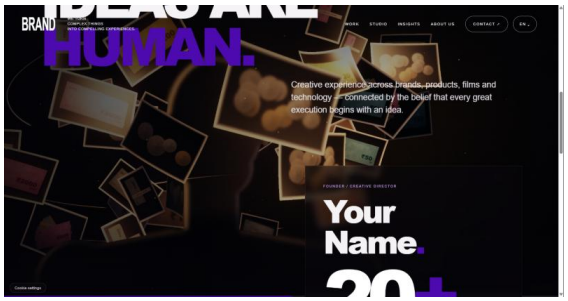
75%: bio, cards around subject

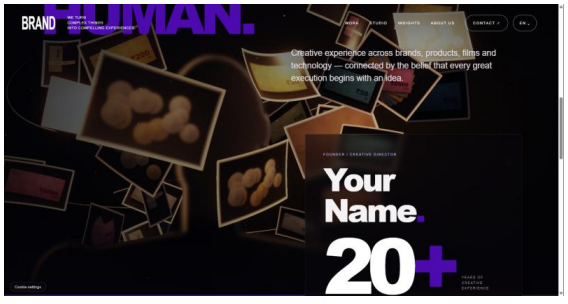


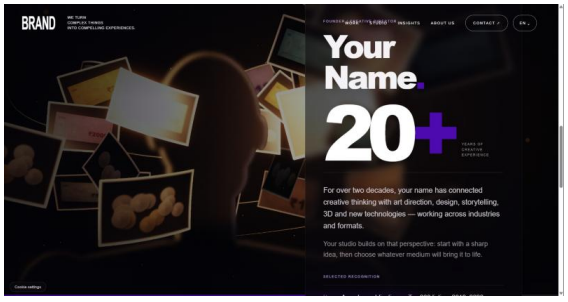
99%: awards + signature row

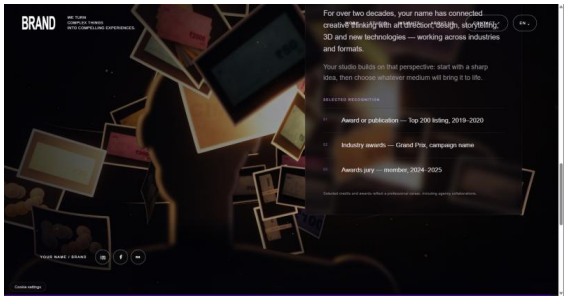
Build











3. How the pieces connect

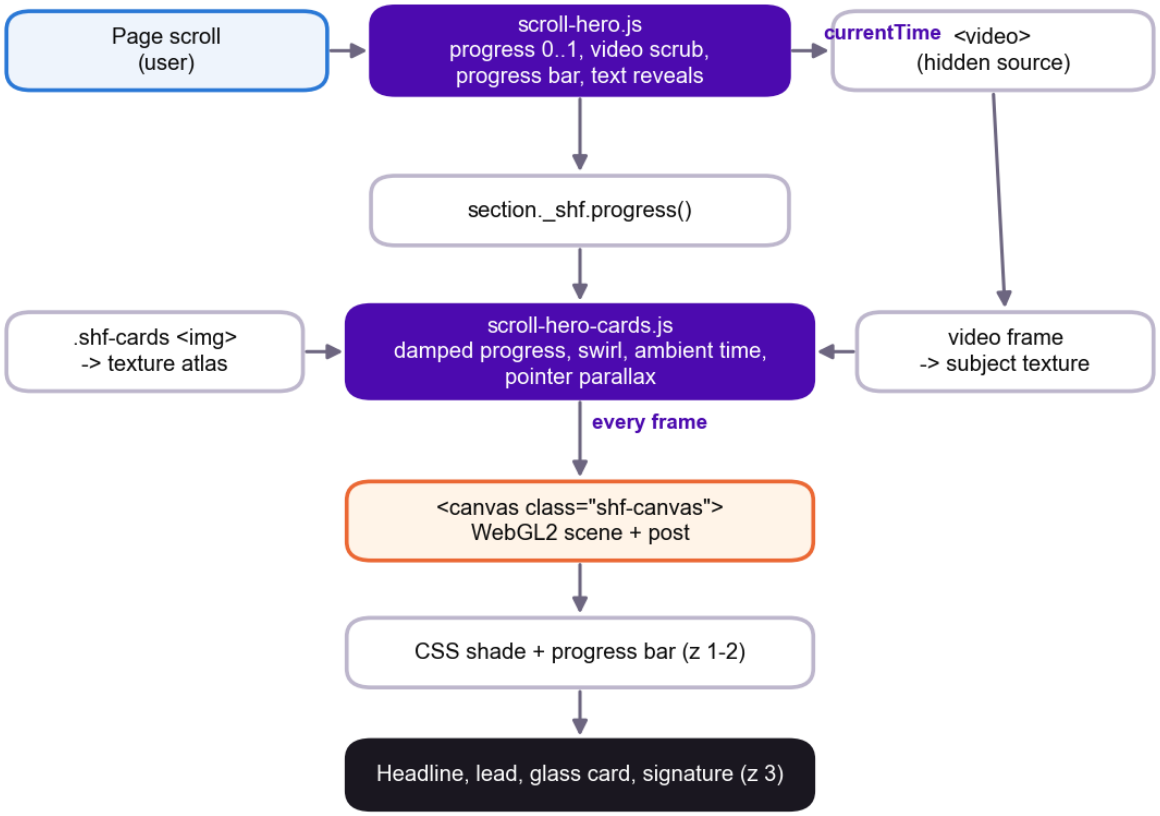


Figure 3.1 The cards engine never measures the page itself; it reads the same progress the video scrub uses, so they can never drift apart.

The <video> element is still the decoder. When WebGL is running, the section gets the class `shf-has-gl`, the video turns invisible (it keeps decoding) and each new frame is copied into the canvas as the subject texture. Without WebGL2 nothing breaks: the plain scrubbed video stays visible.

4. What happens every frame

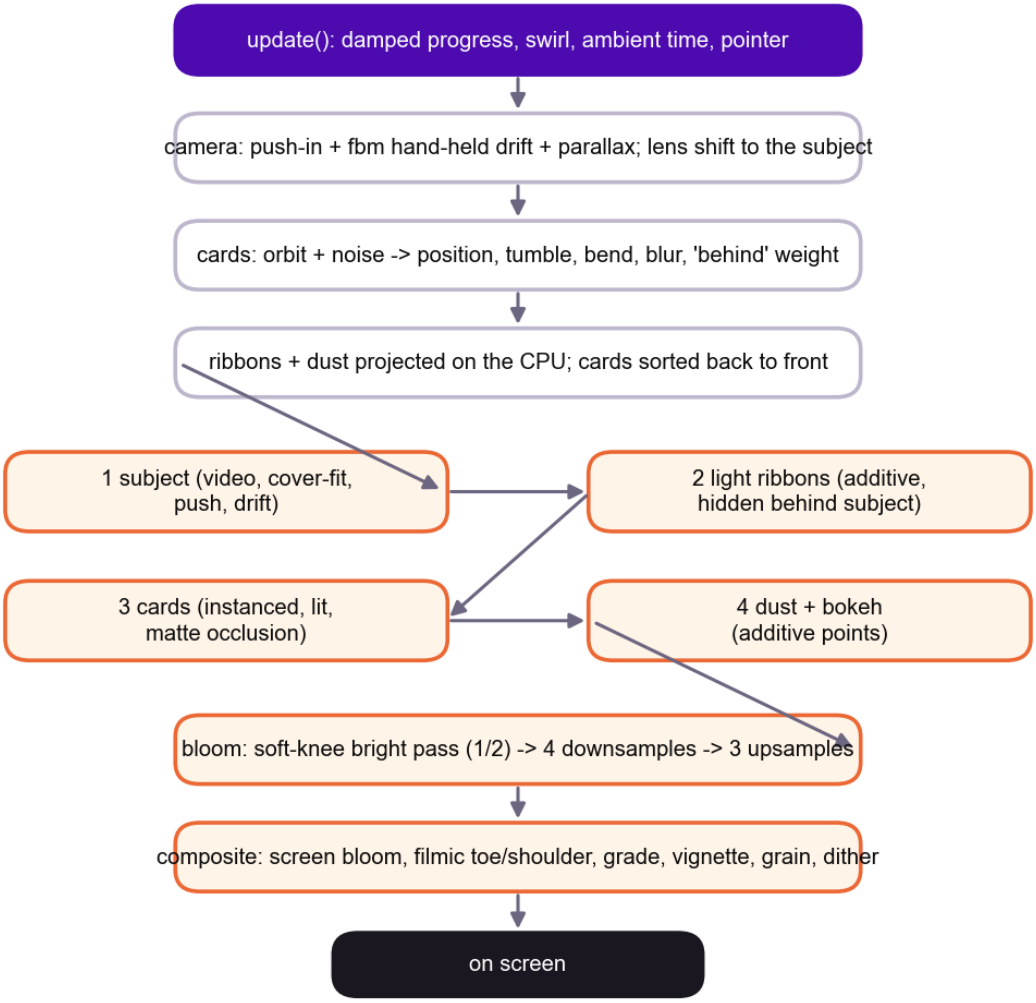


Figure 4.1 One frame. The canvas renders only while the section is on screen, and lowers its own resolution if frames get slow.

5. Non-robotic flow

Rigid motion comes from positions locked directly to the scroll bar. Here three layers break that up:

- **Inertia**: the scene follows a damped copy of the scroll progress (time constant 0.32 s), so it eases into place.
- **Swirl**: the speed of that damped progress pushes the cards a little further around the ring; it decays over 0.6 s, so a quick flick makes the cloud lean and settle like something with weight.
- **Noise everywhere**: each card's radius, height, yaw, pitch, roll and paper bend follow their own fbm curves (THE DAY's value-noise recipe), driven by scroll and by slow ambient time. The camera adds a push-in and a hand-held drift.

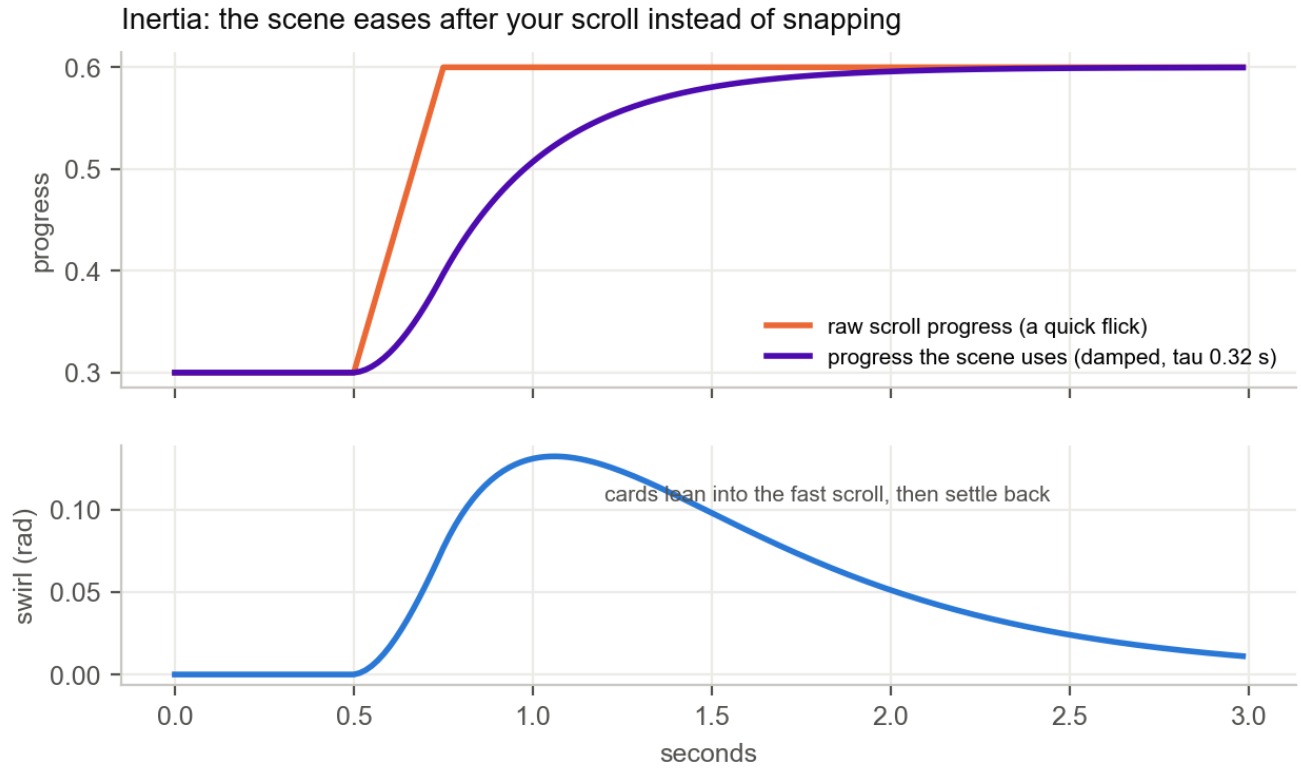


Figure 5.1 A quick scroll flick: the raw input jumps; the scene eases (top) and the swirl overshoots and settles (bottom).

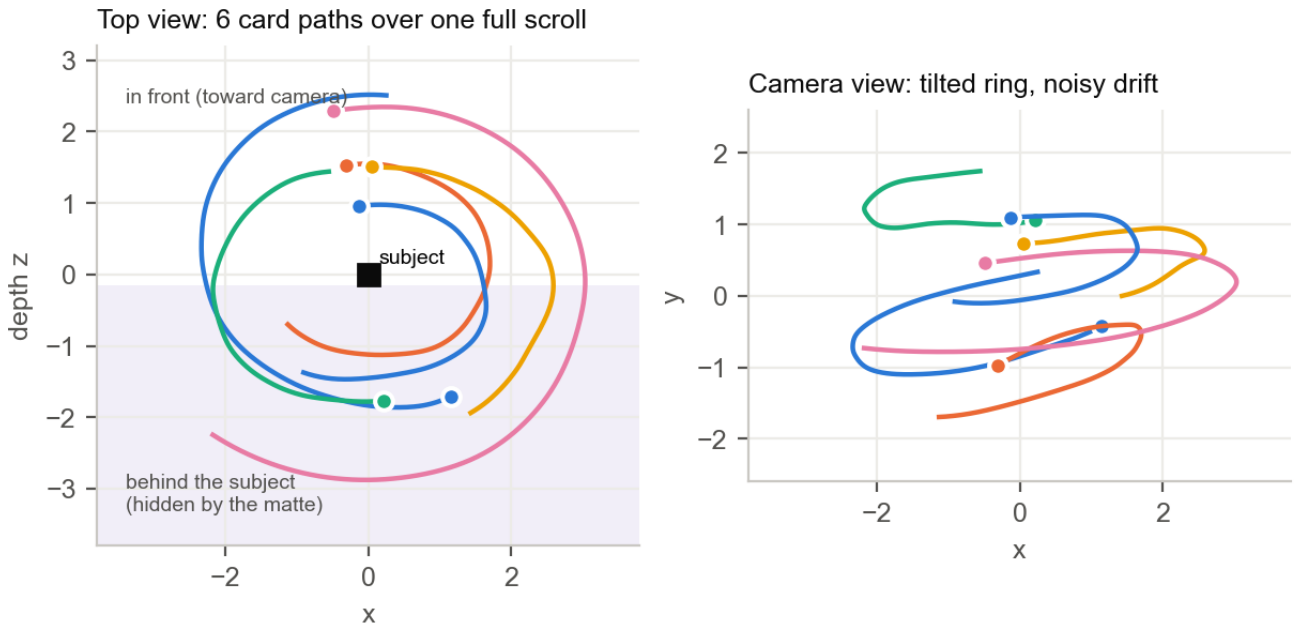


Figure 5.2 Real card paths computed with the engine's formulas. Paths wander instead of tracing circles; the part of the ring behind the subject is masked.

6. Lighting and texture on the cards

Effect	How it is made	What you see
Photo print	rounded-rectangle SDF, 4.5% white border, paper-fibre noise, print grain on the image	real printed photos, not flat rectangles
Key light	warm light from upper right, wrapped diffuse	cards facing the light glow amber
Fill light	cool, dim light from the left	shadow sides stay readable, slightly blue
Gloss + glints	two-lobe specular (sharp + broad), glossier on the photo than on the border	highlights sweep across a card as it turns; bloom turns them into glints
Rim (fresnel)	brightens at grazing angles	edges light up as cards turn side-on
Golden edge glow	emissive band on the card outline, scaled by key-light facing	the gold-lit edges from the reference video
Paper back	warm grey back face	cards flipping over show their backs
Paper flex	card mesh is 14 x 8 quads, bent by noise	cards bow slightly as they fly
Depth of field	texture LOD + 7-tap disc blur by distance from focus	near and far cards go soft
Depth fog	mix toward studio black with distance	far cards sink into the dark
Occlusion	packed matte (or brightness matte) sampled per pixel	cards pass behind the subject
Ribbons + dust	additive light trails and bokeh points, also masked by the subject	golden streaks and floating specks
Finish (D:\day)	bloom chain, filmic toe/shoulder, teal shadows / cream highlights, vignette, grain, dither	filmic glow, rich blacks, no banding

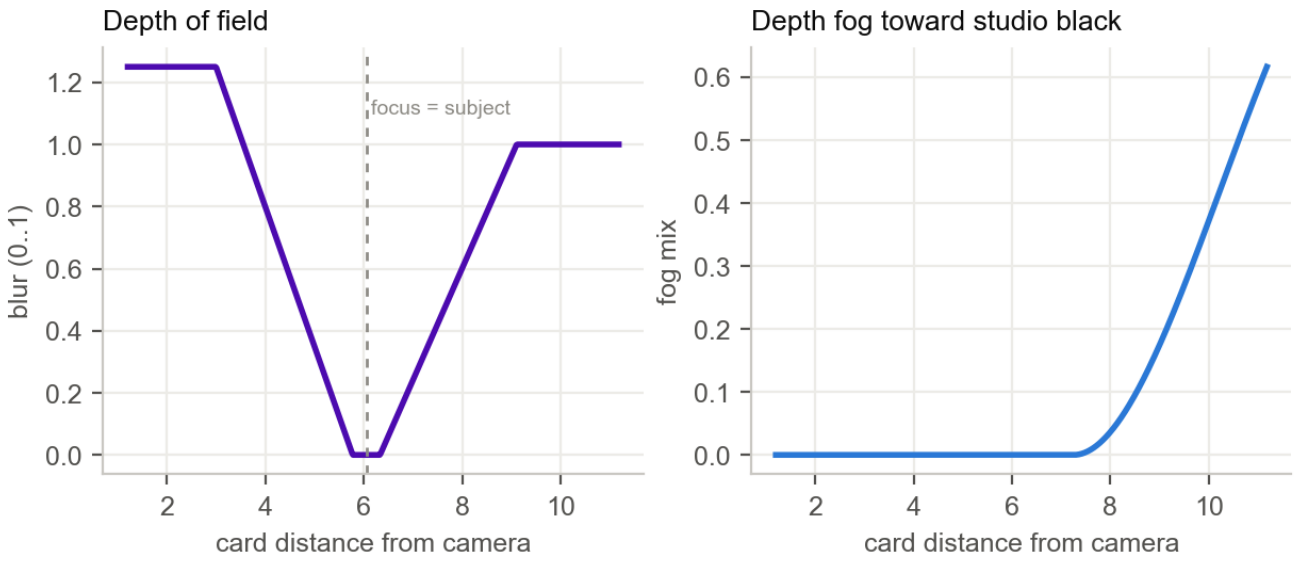


Figure 6.1 Blur and fog as functions of distance. Focus follows the camera as it pushes in.

7. Subject, matte and card images

The placeholder subject clip uses the **packed** format: colour on the left half, the subject's white cut-out on the right half. The engine shows only the colour half and uses the cut-out to hide cards and ribbons that are behind the subject.

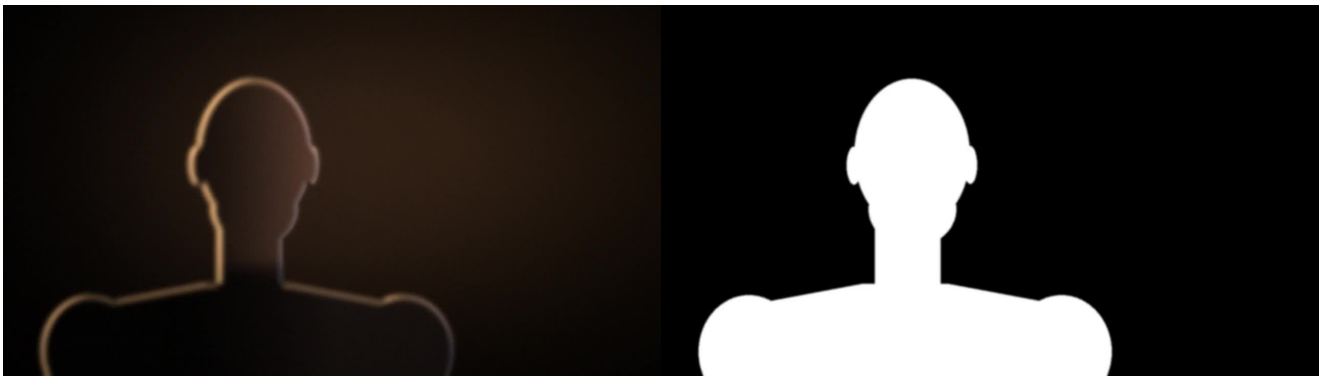


Figure 7.1 One frame of assets/scroll.mp4 (2560 x 720): colour | matte.

With ordinary footage (no matte), leave out `is-packed`; cards behind the subject are then hidden wherever the picture is bright, which suits dark-studio shots like the reference.



Figure 7.2 The 12 placeholder card images (stylised SPECIMEN rupee notes and coins; not reproductions of real notes).

8. Phones

Below 820px the frame stacks: heading, then the video panel (with the cards inside it), then the text. The engine uses about 60% of the cards and a lower pixel ratio there.



Figure 8.1 375 x 812: video panel with 21 cards and light ribbons. No horizontal scroll.

9. Verification

Check	Result
Engine starts, no console errors	pass: WebGL2, 34 cards, 12 image slots, packed subject detected
Scroll position drives video + cards	pass: progress 0.33 / 0.52 / 0.61 / 0.75 / 0.99 rendered as expected (section 2)
Cards pass behind the subject	pass: visible in the 75% and 99% frames
Desktop frame rate (this laptop, 1600x832 at 1.5x)	238 fps measured over 3 s (display-refresh bound), no adaptive downscale triggered
Phone layout 375x812	pass: stacked layout, 21 cards, no horizontal scroll
Fallback without WebGL2	by design (code path): plain scrubbed video; not tested on a device without WebGL2
Real iPhone Safari / Android Chrome	not tested on physical devices
Reduced motion	code path present; not toggled in this test

Not finished / known limits

- The subject is a placeholder silhouette. The look of the final piece depends on your real footage: a 6-8 s continuous shot in a dark studio, ideally with a matte (see prompt file 02).
- The placeholder rupee art is generated; swap in your real images by editing the <img> list.
- The browser pane in this test is a fast desktop GPU; check a mid-range phone before shipping.

Did this need extra services or connectors?

No. Everything was built and tested with tools already on this machine (Python with ffmpeg, Node, and the app's built-in browser). Optional extras that would help later: an AI video tool or a camera session for the real subject footage, a background-removal tool to make its matte, and hosting (for example Vercel, which is already listed in your setup but needs signing in) when you want to share a live link.

Next steps

- Record or generate the subject clip, make a matte, encode as a packed video (README has the command).
- Replace the card images.
- Tune per project with the data-* options (count, subject position, bloom, grain, seed).